① PHYS-3701 Electronics-I B.S 4 year Fall 202

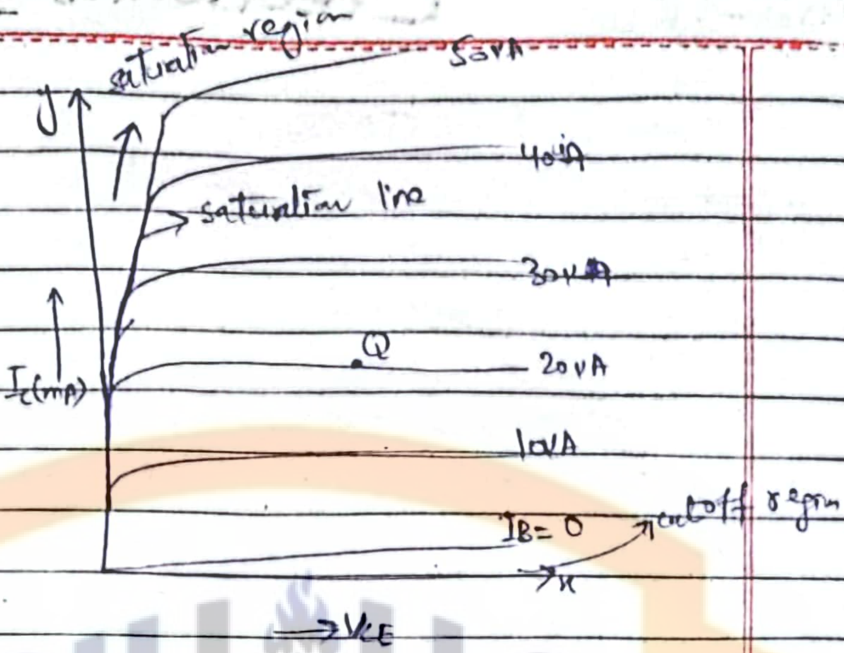
Short Questions:-

(i) Differentiate b/w an ideal diode and a Zener diode?

	Ideal diode	Zener diode
1	An ideal diode only conducts in one direction.	A zener diode can conduct in both directions.
2	It has low doping density.	It has high doping density.
3	It has zero voltage drop in forward direction $V_F = 0$	It has a small forward voltage drop (typically around 0.7V for silicon
4	In reverse bias, the ideal diode has infinite resistance and does not conduct any current	In reverse bias, the zener diode allows current to flow once the reverse voltage reaches the zener voltage
5	An ideal diode does not have a breakdown region because it does not conduct in reverse bias.	The zener diode is designed to operate in breakdown region in reverse bias, where it maintains a stable voltage across it, making it useful for voltage regulation

(ii) What is meant by saturation and cutoff lines on the transistor characteristics curves?

The output characteristics curves of a transistor is shown in fig



When the Transistor operates either at $I_B = 0$ or $I_B < 0$, the region is called cut off region.

The region where we get more value of current is called saturation region and defined as when collector junction as well as emitter junction is forward biased, in this region V_{CE} drops to very small value and collector current is independent of base current

iii) **What is the usual relation b/w A_i and A_v in Transistor amplifiers?**

We know that Voltage gain for Common emitter amplifier is,

$$A_{ve} = \frac{-h_{fe} R}{h_{ie}}$$

Current gain for common emitter amplifier is given by as

$$A_{ie} = h_{fe}$$

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$$\frac{A_{ve}}{A_{ie}} = \frac{-(h_{fe}/h_{ie})R}{h_{fe}}$$

$$\frac{A_{ve}}{A_{ie}} = \frac{-R}{h_{ie}}$$

$$A_{ve} = \frac{-A_{ie}R}{h_{ie}}$$

In general,

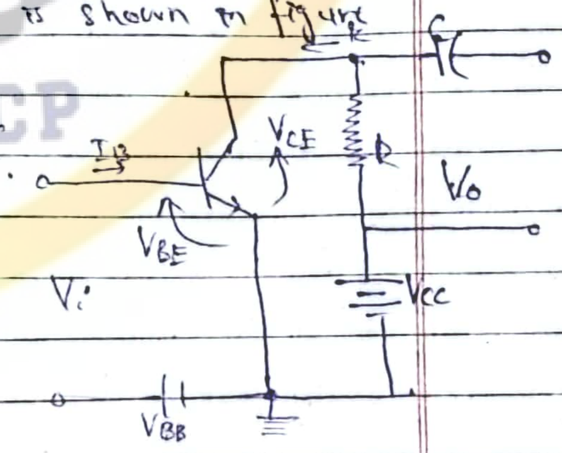
$$A_v = -A_i \frac{R}{R_i} = -A_i \cdot \frac{\text{load resistance}}{\text{input resistance}}$$

This is the required relation between voltage gain and current gain.

(iv) **What is load line? Write down the value of V_{CE} when Q-point is fixed at the midpoint of load line?**

A load line is a line drawn on the current-voltage characteristics graph for a non-linear device like a diode or transistor.

A simple transistor amplifier is shown in figure. Transistor is in series with the resistance R in output circuit. Writing voltage equation for output loop,



$$V_{CC} = V_{CE} + i_C R \Rightarrow i_C R = -V_{CE} + V_{CC}$$

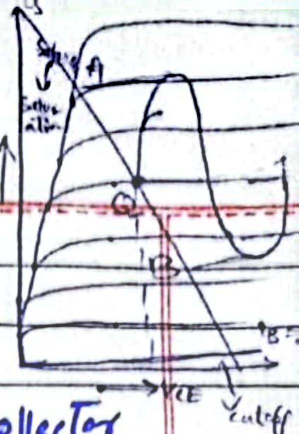
$$i_C = \frac{-V_{CE}}{R} + \frac{V_{CC}}{R} \rightarrow (1)$$

This is equation of straight lines with slope $(-1/R)$ and intercept (V_{CC}/R). This one equation is not sufficient to determine the two unknowns i_C and V_{CE} present in this equation. However a second relation b/w ... characteristics of transistor.

It is indicated by simultaneous solution of equation ① & output characteristics that there is a straight line which represents equation ①. This straight line is called

(4)

DATE __/__/__ load line.



When Q-point is fixed at the midpoint of load line $V_{CE} = \frac{V_{CC}}{2}$

(v) Draw the circuit of a common collector amplifier. Why it is called emitter follower?

From Ilmi Book page # 84.

(vi) What is construction difference between a JFET and MOSFET?

From Ilmi page # 127, 130

From MBD page # 92, 96

Long Questions:-

Q#1(a)

Discuss the circuit operation of a Capacitor (C)- Filter when it is connected to a full-wave center-tapped filter circuit. Derive an expression for its ripple factor (6)

From Ilmi page # 30

(b)

Draw the h-parameter equivalent model circuit for the two-port network. How would you convert an active one port network into its equivalent current source circuit. (4)

From Ilmi page # 58, 60

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Q#2 (a)

Draw the circuit for the common emitter (CE) transistor amplifier and discuss its performance by giving expressions for voltage gain, current gain, input resistance and the output resistance. (6)

From Ilmi page # 81 – 83

(b)

Design Voltage-feedback bias circuit with the following parameters.

$I_C = 2.3\,mA$, $I_B = 30\,\mu A$, $V_{BE} = 0.5\,V$, $h_{FE} = 75$, and $V_{CC} = 15V$.

Use standard assumptions if needed. (4)

From Ilmi page # 114

Q# 3 (a)

Describe how pinch-off is obtained in an n-channel JFET? (5)

From Ilmi page # 127 – 129

(b)

Describe the operation of FET base voltage divider bias amplifier circuit. (5)

From Ilmi page # 134

From floyd 9th edition page # 403.